How to design an arbitrage bot

Arbitrage within Uniswap

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Agenda:

- A living example.
- How to design a flexible arbitrage contract.
- How to search for arbitrage opportunities.

Before getting started, we recommend reader to read documents of uniswap:

- https://uniswapv3book.com/index.html
- https://docs.uniswap.org/

A living example

- `0x53663b0649dd238f1e0434adec853ce405af108595155e950027e2ec9e1ba68a` at block 22185681.
- https://etherscan.io/tx/0x53663b0649dd238f1e0434adec853ce405af108595155e950027e2ec9e1ba68a
- https://app.blocksec.com/explorer/tx/eth/0x53663b0649dd238f1e0434adec853ce405af108595155e950027e2ec9e1ba68a?line=1

Balance changes of this transaction

Balance Changes

Addresses	Token	TokenID	Balance
0x00af7ad1bad0c331d30175cdd461c19709219b10 [Sender]	Ether	-	-0.000000000000000957
0x1409262a7a05c7b1ca7675c12721b18328686022 [Receiver]	Ether	-	+0.000202609345363525
Uniswap V3: BITCOIN	Wrapped Ether	-	-0.158754775792494336
	HarryPotterObamaSonic10Inu: BITCOIN Token	-	+6,907.46543731
Titan Builder	Ether	-	+0.000716396547493442
Uniswap V2: BITCOIN-SPX	SPX6900: SPX Token	-	+633.24318672
	HarryPotterObamaSonic10Inu: BITCOIN Token	-	-6,907.46543731
Uniswap V3: SPX 2	SPX6900: SPX Token	-	-633.24318672
	Wrapped Ether	-	+0.15848066859698304
GLP	Circle: USDC Token	-	+3,428.866681
	Tether: USDT Stablecoin	-	-3,429.14269
Wrapped Ether	Ether	-	-0.000919005892856 01
	Wrapped Ether	-	+0.000919005892856 01
Uniswap V3: USDT 9	Wrapped Ether	-	-1.876609404854217972
	Tether: USDT Stablecoin	-	+3,429.14269
Uniswap V3: USDC 4	Circle: USDC Token	-	-3,428.866681
	Wrapped Ether	-	+1.875964506156873258

A human readable workflow (Part1)

Balance Changes

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0x00af7ad1bad0c331d30175cdd461c19709219b10 [Sender]	Ether	-	-0.000000000000000957
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①Assemble the payload and lunch the tx.

④ `sell` 6907 bitcoin for 0.15875 WETH token

③ `sell` 633 SPX for 6907 bitcoin token

② `sell` 0.15848 WETH for 633 SPX token

A human readable workflow (Part1)

Balance Changes

Addresses	Token	TokenID	Balance
0x00af7ad1bad0c331d30175cdd461c19709219b10 [Sender]	Ether	-	-0.000000000000000957
0x140	Ether	-	+0.000202609345363525
Unisw	Wrapped Ether	-	-0.158754775792494336
	HarryPotterObamaSonic10Inu: BITCOIN Token	-	+6,907.46543731
Titan	Ether	-	+0.000716396547493442
Unisw	SPX6900: SPX Token		+633.24318672
	HarryPotterObamaSonic10Inu: BITCOIN Token	-	-6,907.46543731
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	Tether: USDT Stablecoin	-	+3,429.14269
Uniswap V3: USDC 4	Circle: USDC Token	-	-3,428.866681
	Wrapped Ether	-	+1.875964506156873258

The arbitrator gains profit of 0.158754775792494336 - 0.15848066859698304 =0.000274

However, 0.000274 is not enough for paying the transaction fee (0.000716396547493442)

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③ *`sell` 633 SPX for 6907 bitcoin token*

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A human readable workflow (Part2)

Balance Changes

Addresses	Token	TokenID	Balance
0x00af7ad1bad0c331d30175cdd461c19709219b10 [Sender]	Ether	-	-0.000000000000000957
0x1409262a7a05c7b1ca7675c12721b18328686022 [Receiver]	Ether	-	+0.000202609345363525
Uniswap V3: BITCOIN	Wrapped Ether	-	-0.158754775792494336
	HarryPotterObamaSonic10Inu: BITCOIN Token	-	+6,907.46543731
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	Tether: USDT Stablecoin	-	+3,429.14269
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	Wrapped Ether		+1.875964506156873258

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⑥ *`sell` 3,428.86668 USDC for 3,429.14269 USDT token*

⑦ *`sell` 3,429.14269 USDT for 1.87660 WETH token*

⑤ *`sell` 1.87596 WETH for 3,428.86668 USDC token*

A human readable workflow (Part2)

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	Tether: USDT Stablecoin	-	-3,429.14269
Wrapp	Ether	-	-0.000919005892856 01
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Unisw	Wrapped Ether	-	-1.876609404854217972
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⑤ *`sell` 1.87596 WETH for 3,428.86668 USDC token*

The arbitrator gains profit of 1.876609404854217972 - 1.875964506156873258 = 0.000644898697344631 3

The arbitrator eventually got profit of 0.0006448986973446313 + 0.000274 - 0.000716396547493442 = 0.000203 eth in total.

The price deviation of different pools makes the transaction profitable.

The price of the token of uniswap v2 follows the formula of P_{1} = liquidity of token0 / liquidity of token1. Such that P_{b}^{s} of `Uniswap V2: BITCOIN-SPX` is 47280536047575 / 51798234892358 9 = 0.09127827646217664.

The price of tokens of uniswap v3 is kind of complex, however, we can find it out in `slot0`. By using the formula of Price = (sqrtPriceX96 / 2^96) ^ 2, we got P_{w}^{b} = 232181.66726585582 (`Uniswap V3: BITCOIN`) and P_{s}^{w} = 4.008805707997408e-07 (Uniswap V3: SPX 2).

The P_{s}^{b} derived from pools `Uniswap V3: BITCOIN` and `Uniswap V3: SPX 2` is P_{s}^{b} = P_{w}^{b} * P_{s}^{w} = 0.09307711930277178.

We conclude the SPX token is more cheaper by trading against the joint of `Uniswap V3: BITCOIN` and `Uniswap V3: SPX 2` pools than `Uniswap V2: BITCOIN-SPX` pool. This is why the arbitrator by SPX from `Uniswap V3: BITCOIN` and sell them to the counterpart.

The actual workflow

- From Uniswap V3: BITCOIN To 0x1409262A...328686022 For 0.158754775792494336 ($290.40) Wrapped Ethe... (WETH)
- From Uniswap V3: SPX 2 To Uniswap V2: BITCOIN-SPX For 633.24318672 ($321.94) SPX6900 (SPX)
- From 0x1409262A...328686022 To Uniswap V3: SPX 2 For 0.15848066859698304 ($289.90) Wrapped Ethe... (WETH)
- From Uniswap V2: BITCOIN-SPX To Uniswap V3: BITCOIN For 6,907.46543731 ($302.75) HarryPotterO... (BITCOI...)
- From Uniswap V3: USDT 9 To 0x1409262A...328686022 For 1.876609404854217972 ($3,432.80) Wrapped Ethe... (WETH)
- From Uniswap V3: USDC 4 To 0xB9A44069...3A040152e For 3,428.866681 ($3,428.56) USDC (USDC)
- From 0x1409262A...328686022 To Uniswap V3: USDC 4 For 1.875964506156873258 ($3,431.62) Wrapped Ethe... (WETH)
- From 0xB9A44069...3A040152e To Uniswap V3: USDT 9 For 3,429.14269 ($3,428.43) Tether USD (USDT)

The actual workflow is hard to read, for the trading path is optimized for saving gas.

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Arbitrator sell BITCOIN for WETH token, after receiving the WETH token, the arbitrage contract shall send the BITCOIN token to the pool of `Uniswap V3: BITCOIN` in it's `uniswapV3SwapCallback` function.

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In order to get the BITCOIN token for the pool of `Uniswap V3: BITCOIN`, the arbitrage contract turns to pool of `Uniswap V3: SPX 2` to get SPX token, by setting the parameter of `tokenOut` to the pool of `Uniswap V2: BITCOIN-SPX`, the bought SXP token is send to that pool directly.

The actual workflow

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`Uniswap V3: SPX 2` pool ask the arbitrage contract to send the WETH to finish this trade (by call the `uniswapV3SwapCallback` function within the arbitrage contract), and the arbitrage contract do it this time.

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Uniswap v2 works differently from v3, that is `tokens must be transferred to pairs before swap is called`. Since SPX token is send to the `Uniswap V2: BITCOIN-SPX` pool previously (step 2), the arbitrage contract lunch the swap and the required BITCOIN token (step 1) is send to the `Uniswap V3: BITCOIN` pool.

At last, all pools are satisfied with no revert!

How to design a flexible arbitrage contract (TODO)

How to search for arbitrage opportunities (TODO)